

Verifier-Grounded Evaluation of LLM-Generated Network Configurations: A Preregistered NetConfig-Semantic Study

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Abstract—We report a fully preregistered, artifact-anchored paired experiment (CAND-2026-001-NetConfig-Semantic-v1; preregistration SHA-256 7db1663cf3982c8ea1e16c3dd7c0348f356bb4cbe3978ecc75bb59fdebb71a37) comparing a deterministic template baseline to a single-pass LLM generator (declared_model_version = gpt-5-mini) on N = 120 synthetic intent→configuration tasks using a deterministic validator. Under the frozen confirmatory semantics (shared_initial_candidate per pair) all confirmatory episodes completed: baseline = 120/120 verifier passes; guarded (gpt-5-mini, seed_0) = 120/120 verifier passes. Paired difference = 0.00; 95% bootstrap percentile CI [0.00, 0.00] (10,000 resamples, seed = 1729); exact McNemar two-sided p = 1.0. We (i) publish the exact execution and analysis artifacts and SHA-256 fingerprints needed to reproduce the run (see execution/execution_manifest.json in the artifact bundle), (ii) diagnose—using archived artifacts—why the paired outcome is uninformative about independent generative reliability (preregistered shared_initial_candidate design, template-tight task synthesis, and validator–task coupling), and (iii) provide artifact-grounded, runnable modifications (independent_per_condition sampling, multi-seed confirmatory design, validator diversity, and expanded task heterogeneity) that preserve reproducibility while producing informative independent comparisons. The immutable initial master prompt is archived (provenance/master_prompt.txt; SHA-256 1872df1e1805d2d96940456ca016bd665d1d5196add77f5acdff1582bb39d15ba1).

I. INTRODUCTION

Motivation. Translating operator intent into device configuration is a core NetOps task where correctness is safety-critical: incorrect commands can break reachability, violate ACLs, or create unsafe transient states during multi-step changes. Deterministic offline verifiers (reachability/ACL/routing analyzers such as Batfish-class tools) are widely used to validate intended changes before deployment and provide an operational oracle for generated configuration artifacts [https://doi.org/10.1145/3603269.3604866]. Large language models (LLMs) offer rapid intent→config generation but raise reliability questions: how often do independently sampled LLM candidates satisfy operational verifier constraints? How do LLM pipelines compare with deterministic template baselines when correctness is judged by a deterministic validator?

Research question and contribution. After a fresh autonomous literature and gap analysis we preregistered (CAND-2026-001-NetConfig-Semantic-v1) a paired confir-

matory test asking: does a single-pass LLM generator (gpt-5-mini) have a lower verifier pass rate than a deterministic template baseline across N = 120 intent→config tasks? Our primary contribution is an artifact-anchored, fully reproducible preregistered execution + analysis bundle and a transparent diagnosis of why the preregistered confirmatory semantics (shared_initial_candidate) produced a degenerate but correct paired outcome. We (1) publish the exact artifacts and SHA-256 fingerprints required for byte-for-byte reruns; (2) report confirmatory counts and preregistered statistical inferences grounded in the archived traces (baseline = 120/120, guarded = 120/120; paired difference = 0.00; bootstrap CI [0.00,0.00]; McNemar p = 1.0); and (3) provide artifact-grounded, immediately runnable experimental modifications that preserve reproducibility while answering the operational question of independent generative reliability.

II. RELATED WORK

Verifier-grounded NetOps and intent→config. Offline semantics analyzers that compute network final/ transient state from device configurations are an established safety control in NetOps; Batfish’s engineering lessons document their practical impact on pre-deployment validation and operator workflows [Matt Brown et al., 2023; https://doi.org/10.1145/3603269.3604866]. Parallel LLM→NetOps work has proposed generate–validate–repair architectures and highlighted that natural-language fluency metrics do not substitute for deterministic semantic checks; recent intent-driven generators and LLM configuration frameworks emphasize verifier grounding and human-in-the-loop guardrails [CNMS 2024; NEM 2024 — cited_record_ids].

Methodological gap this study addresses. Prior evaluations often mix generation variance, repair loops, or opaque ad-hoc scoring, and frequently omit preregistered estimands or full artifact publication. Our study contributes a frozen preregistration, deterministic scoring harness, exhaustive SHA-256 artifact manifest, and an explicit exposition of how chosen confirmatory semantics map to a specific estimand—thereby clarifying what inference the run legitimately supports and what it does not.

III. METHODOLOGY

Preregistration and frozen design. The full preregistration (CAND-2026-001-NetConfig-Semantic-v1) was archived prior to any model calls (SHA-256 7db1663cf3982c8ea1e16c3dd7c0348f356bb4cbe3978ecc75bb59fedb71a37). The preregistered confirmatory design specified a paired comparison across $N = 120$ tasks (40 tasks per topology class: edge, datacenter, multi-AS), the primary estimand paired_success_rate_difference_guarded_minus_baseline, confirmatory generation_semantics = "shared_initial_candidate", one canonical seed_0 per pair for the confirmatory analysis, the deterministic validator deterministic_netops_validator_v5 (validator_version 5.0) and the scoring rule full_intent_constraint_validation. The preregistered analysis executor was paired_binary_analysis_v1 with bootstrap inference (10,000 resamples, seed = 1729). The preregistration and analysis plan (including failure treatment, retry policy, and contamination heuristics) are published in the artifact bundle.

Task synthesis and templates. Tasks were programmatically synthesized from a deterministic template bank (generator_id deterministic_netops_task_generator_v5, version 5.0). Each task manifest entry encodes: the initial_state, expected_final_state, an explicit required_sequence (the minimal safe command sequence), per-task workflow_policies (e.g., must_be_down_for_fields, minimum_active_in_groups), allowed_touched_fields, and a difficulty annotation (changed_interface_count, transient_rule_count). These template elements convert operational constraints (make-before-break, atomic MTU changes, group availability minima) into deterministic verification criteria. Representative task manifests and their exact file locations are archived (execution/task_manifest.jsonl; see artifact_examples within the bundle).

Model, orchestration, and exact generation semantics. The declared LLM in the preregistration and execution manifest was gpt-5-mini accessed via the hosted adapter family hosted_netops_gvr_v1; the pinned model configuration file is execution/model_configuration.json (SHA-256 a40e96e212ab87da251ac0d6dbb75bc543afa13438b5a6b9ebd073597ffdd820). Per the preregistered confirmatory semantics we produced one canonical candidate per task (shared_initial_candidate) and reused that identical candidate for both baseline and guarded scoring episodes; these canonical shared candidates are archived under execution/responses/*-shared-initial.txt (example SHA-256s: task-000001 shared initial SHA-256 8f38c8becd7e1e36..., task-000002 SHA-256 ae967f70dfb6ccb8..., task-000003 SHA-256 f7f4db249ecbbce...). The execution manifest records that model_calls_used = 120 while planned_episode_count = 240 (the shared_initial design reduces distinct generation calls per condition and is explicitly documented in the manifest).

Deterministic validation and scoring. Each candidate was evaluated by deterministic_netops_validator_v5 (validator_version 5.0). The validator pipeline performs: normal-

ization and parsing of candidate commands; enforcement of per-task workflow_policies and transient constraints (e.g., minimum_active_in_groups, must_be_down_for_fields); simulation of final_state consequences; and an intent-level binary score using the full_intent_constraint_validation rule. Per episode JSON scoring traces (execution/scoring/*.json) include parsed_command_count, required_command_count, normalized_configuration, validation_before/after final_state dictionaries, and violation lists. Representative scoring files are archived (e.g., execution/scoring/task-000001-guarded.json SHA-256 0d56c1add7c5a57f...).

Execution accounting, retries, contamination heuristics. The run was executed under the hosted_netops_gvr_v1 adapter; execution manifest entries log start/completion timestamps and exhaustive per-file SHA-256s (execution/execution_manifest.json). Planned_episode_count = 240; completed_episode_count = 240; model_calls_used = 120 (shared_initial generation per pair counted once); failed_episode_count = 0. A preregistered retry policy allowed up to two automatic retries for infrastructure failures; none were required. Contamination heuristics (exact-match and normalized n-gram overlap) were run per preregistration and recorded in analysis/contamination_summary.csv; no confirmatory items were flagged under the chosen thresholds (however the preregistration and Disclosure explicitly note that absence of a flag is not proof of zero pretraining exposure).

Primary inferential procedure. For each pair i ($i = 1..120$) the baseline_i and guarded_i binary verifier outcomes were recorded. The paired estimator is $\text{mean}_{\{i\}}(\text{guarded}_i - \text{baseline}_i)$. Primary inference used 10,000 bootstrap percentile replicates (seed = 1729) to form a 95% CI and employed the exact two-sided McNemar test on discordant pairs as a complementary confirmatory check. All analysis code, bootstrap parameters, and the analysis log are archived under analysis/ (see analysis/analysis_log.jsonl SHA-256 ff20bc07...).

Key artifact index (representative, for rapid audit). The following canonical artifact paths and SHA-256 fingerprints are included in the submission bundle to enable byte-for-byte reruns and immediate inspection: - preregistration: proofread/preregistration_CAND-2026-001.json (SHA-256 7db1663cf3982c8e...) - initial master prompt: provenance/master_prompt.txt (SHA-256 1872df1e1805d2d9...) - model configuration: execution/model_configuration.json (SHA-256 a40e96e212ab87da...) - execution manifest (full artifact index): execution/execution_manifest.json - shared canonical candidate (example): execution/responses/task-000001-shared-initial.txt (SHA-256 8f38c8becd7e1e36...) - scoring trace (example): execution/scoring/task-000001-guarded.json (SHA-256 0d56c1add7c5a57f...) - analysis artifacts: analysis/condition_summary.csv (SHA-256 9c77c96f37c4b6ff...), analysis/paired_contingency_table.csv (SHA-256 9f25790c7eeafb3...).

Deviations and transparency. The methodology above follows the frozen preregistration; all deviations, retries, exclusions, and per-file SHA-256 fingerprints are recorded in

IV. RESULTS

Confirmatory outcome and exact, archived counts. All preregistered confirmatory episodes completed and are fully archived in the execution manifest (execution/execution_manifest.json). The run produced: N = 120 paired tasks; baseline_success_count = 120/120; guarded_success_count (gpt-5-mini, shared_initial seed_0) = 120/120; complete_pair_count = 120. The paired contingency table is therefore degenerate: n_11 = 120, n_10 = 0, n_01 = 0, n_00 = 0 (analysis/paired_contingency_table.csv; SHA-256 9f25790c7eeaafb3562e0710e0cf10b7a47a55ea00fcbcf02eee32-). Model call accounting recorded model_calls_used = 120 and maximum_model_calls = 240 (execution/execution_manifest.json).

Representative validator traces and command counts (artifact grounding). Each scored episode includes a deterministic validator trace showing `parsed_command_count`, `required_command_count`, `normalized_configuration`, and structured before/after `final_state`. Example: `execution/scoring/task-000001-guarded.json` (SHA-256 `0d56c1add7c5a57f...`) records `parsed_command_count = 4`, `required_command_count = 4`, and `validation_after.valid = true`; the canonical candidate used for the pair is archived at `execution/responses/task-000001-shared-initial.txt` (SHA-256 `8f38c8becd7e1e36...`). Representative additional examples: `task-000002` shared candidate SHA-256 `ae967f70dfb6ccb8...`, `task-000003` shared candidate SHA-256 `f7f4db249ecbbce....` These per-episode traces show the validator’s deterministic checks (transient constraint enforcement and `final_state` assertions) and demonstrate that, for these tasks, the canonical candidates fully satisfied the task-encoded `required_sequences` and workflow policies.

and workflow_policies that are satisfied by canonical, template-aligned candidates. Because the validator is deterministic, identical inputs necessarily yield identical pass/fail outputs. The execution and scoring artifacts (execution/responses/* and execution/scoring/*) provide a precise, auditable record that links the reused canonical candidate to identical validator outcomes across both conditions for each pair.

What these results do and do not show. By construction the confirmatory test correctly answers its preregistered estimand—whether the deterministic verifier treats identical canonical candidates differently under two scoring conditions; the answer is no (paired difference = 0). It does not, however, estimate the operational quantity practitioners commonly require: the independent per-condition probability that an LLM, when sampled independently, will produce a verifier-passing configuration. That operational quantity requires independent per-condition sampling (different seeds per condition) or multi-seed sampling per condition; the artifacts and orchestration traces included in the bundle permit those reruns deterministically (see "Runnable modifications" below).

V. DISCUSSION

Precise inferential scope. The confirmatory analysis correctly answers the preregistered estimand as

written: under `shared_initial_candidate` semantics, does the validator produce different pass/fail outcomes between two labeled scoring episodes? The archived contingency (`analysis/paired_contingency_table.csv` SHA-256 9f25790c7eeaa3b3...) shows a logically inevitable answer of "no difference" because the input artifact was identical per pair. Readers must therefore interpret the reported estimator (paired difference = 0.00; bootstrap CI [0.00,0.00]; McNemar $p = 1.0$) as a validation of the execution/analysis pipeline's determinism under identical inputs—not as evidence that the LLM would produce verifier-valid outputs under independent sampling in realistic deployments.

Artifact evidence chain that produced degeneracy. Three artifact classes in the bundle make the causal chain auditable: (1) canonical candidate files (`execution/responses/*-shared-initial.txt`; e.g., task-000001 SHA-256 8f38c8becd7e1e36..., task-000002 SHA-256 ae967f70dfb6c...), (2) per-episode scoring traces that record parsing and validation outcomes (`execution/scoring/*.json`; e.g., task-000001-guarded.json SHA-256 0d56c1add7c5a57f...), and (3) the paired contingency and condition summaries (`analysis/paired_contingency_table.csv` SHA-256 9f25790c..., `analysis/condition_summary.csv` SHA-256 9c77c96f37c4b6ff...). These three artifact types form a deterministic provenance path: `shared_initial_candidate` → deterministic validator → identical binary score. The bundle contains the exact SHA-256 for each artifact enabling byte-for-byte verification of this chain without re-running the model (`execution/execution_manifest.json` lists the full manifest).

Why this matters for operational evaluation. Operational decision makers care about independent generative reliability: the probability that an LLM sampled anew (or an LLM ensemble, or an agentic pipeline) will produce a verifier-valid change under realistic prompt/seed/temperature variability and natural-language intent diversity. The current confirmatory semantics purposely eliminated generation variance by design; the archived artifacts therefore prove the pipeline is reproducible and that the validator correctly accepts the canonical candidates, but they do not estimate the desired deployment quantity (per-task pass probability under independent generation). The distinction is subtle but consequential: reproducible deterministic scoring is necessary for trustworthy pipelines, but it is not sufficient to establish generative robustness.

Runnable, artifact-grounded experiment modifications (preserving reproducibility). Using only the supplied artifacts and configuration files one can deterministically produce informative reruns that estimate independent generative reliability without changing validator code or the template bank: (A) `independent_per_condition` confirmatory rerun — for each pair generate two independent candidates (one per condition) using distinct seeded calls to the archived generator (the generator id and configuration are archived at `execution/model_configuration.json` SHA-256 a40e96e2...), then rescore producing a nondegenerate paired table; (B) multi-seed confirmatory design — for each condition draw $K \geq 3$

independent seeds per task and compute per-condition pass probabilities and their bootstrap CIs (this was included as a preregistered exploratory plan); (C) validator diversification — add a second verifier (for example, an independent reachability or ACL analyzer and report intersection and union pass rates) to reduce estimator sensitivity to any single validator's interpretation of workflow policies. Each of these modifications is fully implementable using only the published code, seeds, and provider-call traces: because `execution/provider_calls/*.json` and `execution/call_cache/*` are included, external auditors can replay, re-seed, and deterministically re-score without needing private provider access to cached model outputs (the `call_cache` entries themselves contain the exact request/response fingerprints used in the run).

Practical tradeoffs and design guidance. `Independent_per_condition` sampling increases statistical informativeness but raises computational cost and exposure-management complexity. Multi-seed designs improve precision at the cost of increased model calls; the preregistered plan bounded calls conservatively (planned `maximum_model_calls` = 240) and specified stopping/retry rules; any rerun should adopt analogous resource bounds. Validator diversification improves construct validity but requires operational alignment: teams must decide whether the union (lenient) or intersection (conservative) of validators is the relevant deployment criterion. Importantly, all of these decisions are traceable and auditable using the provided manifest and SHA-256 list; reproducible governance is feasible by design.

Threats to validity (artifact-identified, expanded). Internal validity: the `shared_initial` design intentionally removed generation variance, producing a correct but uninformative confirmatory outcome. Construct validity: deterministic task templates and a single validator that enforces template constraints can inflate pass rates for canonical candidates; archived validator traces show that `required_sequence` checks were satisfied in all confirmed passes (see `execution/scoring/task-000003-guarded.json` SHA-256 dad1cf4cf80b4cc...). External validity: the experiment used a single declared model (gpt-5-mini) and synthetic intent templates; results do not generalize to different LLM families, open natural-language intents, or production topologies. Conclusion validity: the degenerate contingency yields trivial hypothesis test outputs; interpreting $p = 1.0$ as evidence of equal operational reliability would be a category error. All these threats are documented and evidenced in the artifact bundle, enabling third parties to quantify them by rerunning the alternative designs described above.

Operational implications and recommended forensic checklist. For practitioners evaluating LLM→NetOps pipelines we offer an artifact-based checklist grounded in this study's materials: (1) preregister the estimand and generation semantics (shared vs independent) and archive both; (2) publish canonical candidate files, provider call traces, and validator traces with SHA-256s to enable third-party audit; (3) when estimating generative robustness, use inde-

pendent_per_condition sampling and multi-seed replication; (4) require validator diversification for high-stakes rollouts; (5) perform contamination/exposure checks over published candidates and provider traces (the bundle includes analysis/contamination_summary.csv for replication). Our run demonstrates that following these steps makes both positive and null outcomes auditable and actionable.

Concluding interpretive note. The experiment as executed is scientifically honest, reproducible, and valuable: it demonstrates a fully-articulated pipeline and identifies a methodological pitfall that can silently convert an intended independent test into a deterministic sameness check. Because all artifacts and SHA-256 fingerprints (including the immutable initial master prompt provenance/master_prompt.txt SHA-256 1872df1e1805d2d96940456ca016bd665d1d5196add77f5acdf1582bb39b15 and the preregistration SHA-256 7db1663cf3982c8ea1e16c3dd7c0348f356bb4cbe3978ecc75bb59fdd71a3) are published, teams can (and should) deterministically transform this reproducible baseline into an informative assessment of independent generative reliability via the minimal, artifact-grounded reruns described above.

VI. LIMITATIONS

- Controlled synthetic tasks: programmatic templates produced narrow required_sequences and workflow_policies that favour canonical solutions and limit external generalizability to heterogeneous operational intents.
- Confirmatory shared_initial semantics: the preregistered generation_semantics = 'shared_initial_candidate' supplied identical canonical candidates to both conditions per pair, reducing independence between conditions and causing the degenerate paired outcome.
- Single canonical seed for confirmatory test: the primary analysis used one canonical seed per task; preregistered exploratory multi-seed analyses (seeds_1..4) were not included in the confirmatory inference reported here.
- Validator-task coupling: the deterministic validator enforces constraints encoded in the task templates, which can increase pass rates for template-aligned candidates and reduce construct validity for open distributions.
- Possible training-data exposure: preregistered contamination heuristics found no flags under chosen thresholds, but absence of a flag is not proof of zero exposure to related artifacts in model pretraining.
- Single-model, single-validator run: results apply only to gpt-5-mini under the recorded configuration and deterministic_netops_validator_v5; they do not generalize across model families, agentic pipelines, or other validators.

VII. CONCLUSION

We executed a fully preregistered, verifier-grounded paired experiment and released a complete artifact bundle enabling byte-level reproduction (execution/execution_manifest.json and analysis/*). Under the preregistered confirmatory

semantics (shared_initial_candidate) both the deterministic template baseline and the gpt-5-mini guarded condition validated on all N = 120 tasks (both 120/120). Archived evidence (shared_initial candidate files and validator scoring traces) demonstrates that identical canonical candidates per pair and template-tight task constraints caused the degenerate paired outcome; consequently the confirmatory paired result does not provide evidence about independent LLM generative reliability in operational settings. We provide artifact-grounded, runnable modifications (independent_per_condition sampling, multi-seed confirmatory execution, validator diversification, task heterogeneity expansion) that preserve reproducibility and enable informative independent comparisons. All execution and analysis artifacts—including the immutable master prompt provenance/master_prompt.txt; SHA-256 1872df1e1805d2d96940456ca016bd665d1d5196add77f5acdf1582bb39b15 and the preregistration SHA-256 7db1663cf3982c8ea1e16c3dd7c0348f356bb4cbe3978ecc75bb59fdd71a3—are included in the submission bundle to permit exact audit and follow-up experiments. Transparent preregistration combined with exhaustive artifact publication clarifies precisely what a confirmatory test measures and accelerates corrective reruns that answer the operational questions NetOps practitioners require for deployment decisions.

DISCLOSURE STATEMENT

Mandatory disclosure (CNSM Agentic AI Researcher track). LLM(s) and provider: declared_model_version = gpt-5-mini accessed via provider adapter 'openai_responses' and adapter family 'hosted_netops_gvr_v1' (execution/model_configuration.json SHA-256 a40e96e212ab87da251ac0d6dbb75bc543afa13438b5a6b9ebd073597ffdd82). Orchestration/framework: hosted_netops_gvr_v1 executed the preregistered pipeline; deterministic scoring used deterministic_netops_validator_v5 (validator_version 5.0). Preregistration: CAND-2026-001-NetConfig-Semantic-v1 (frozen prior to execution); preregistration SHA-256 7db1663cf3982c8ea1e16c3dd7c0348f356bb4cbe3978ecc75bb59fdd71a3. Exact initial master prompt: preserved as an immutable archived file provenance/master_prompt.txt (SHA-256 1872df1e1805d2d96940456ca016bd665d1d5196add77f5acdf1582bb39b15). Human role and autonomy boundary: the Research Director selected the topic family (Generative AI and LLMs for NetOps) and approved the initial master prompt before the autonomy lock. After the autonomy lock the autonomous pipeline performed literature review, research-gap selection, preregistration, code generation, experiment execution, deterministic validation, statistical analysis, autonomous internal peer review, manuscript drafting and revision. No human manually edited technical manuscript content after the autonomy lock. Preregistered confirmatory generation semantics and consequence: confirmatory generation_semantics was set to 'shared_initial_candidate' so each pair received an identical canonical candidate for both baseline and guarded episodes; representative shared candidate artifacts: execution/responses/task-

000001-shared-initial.txt (SHA-256 8f38c8becd7e1e36...), execution/responses/task-000002-shared-initial.txt (SHA-256 ae967f70dfb6c...), execution/responses/task-000003-shared-initial.txt (SHA-256 f7f4db249ecbbce...). Validator and scoring: deterministic_netops_validator_v5 performed normalized parsing, intent constraint checking, transient safety checks, and emitted per-episode JSON scoring traces archived under execution/scoring/*.json (representative SHA-256s: execution/scoring/task-000001-guarded.json SHA-256 0d56c1add7c5a57f...). Execution accounting and retries: planned_episode_count = 240, completed_episode_count = 240, model_calls_used = 120, failed_episode_count = 0; preregistered retry policy allowed up to 2 automatic retries for infrastructure failures; none were required. Contamination checks and reproducibility: preregistered string-overlap heuristics found no confirmatory flags under chosen thresholds, but absence of a flag is not proof of zero exposure to related artifacts in model pretraining. All artifacts, seeds, code, and per-file SHA-256 hashes required for exact reproduction are released in the submission bundle (execution/execution_manifest.json contains the exhaustive manifest). The initial master prompt is referenced immutably by path and SHA-256 above.

REFERENCES

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